

A Machine-Verified Disproof of TLMC Conjecture #1556: The Chromatic Number of $G(\mathbb{Z}^2, \{1, 2, 4\})$

Batch 9 solution submission

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Abstract

We disprove a conjecture of the Last Math Competition asserting that the chromatic number of the planar distance graph $G(\mathbb{Z}^2, D)$ with $D = \{1, 2, 4\}$ equals 7. In fact $\chi = 3$: the map $c(x, y) = (x + y) \bmod 3$ is a proper 3-coloring, and the axis triangle $\{(0, 0), (1, 0), (2, 0)\}$ is a clique. The coloring is verified in Lean 4 by a mod-3 argument: lattice vectors of squared Euclidean length 1, 4 or 16 have coordinate sum not divisible by 3, since squares are 0 or 1 (mod 3) while $2x^2 \not\equiv 1 \pmod{3}$.

1 The conjecture

Conjecture 1.1 (TLMC-00000001556). The chromatic number of the planar distance graph $G(\mathbb{Z}^2, D)$ with $D = \{1, 2, 4\}$ is 7.

2 The disproof

Theorem 2.1. *Conjecture 1.1 is **false**: $\chi(G(\mathbb{Z}^2, \{1, 2, 4\})) = 3$.*

Proof. Upper bound. Define $c(x, y) = (x + y) \bmod 3$. Two lattice points at Euclidean distance $d \in \{1, 2, 4\}$ differ by a vector (a, b) with $a^2 + b^2 = d^2 \in \{1, 4, 16\}$. Suppose $3 \mid (a + b)$; then $b \equiv -a \pmod{3}$, so

$$a^2 + b^2 \equiv 2a^2 \pmod{3},$$

and $2a^2 \in \{0, 2\} \pmod{3}$ since squares are 0 or 1 (mod 3). But $a^2 + b^2 \equiv d^2 \equiv 1 \pmod{3}$ (as $1, 4, 16 \equiv 1$), a contradiction. Hence the color changes along every edge, and c is a proper 3-coloring.

Lower bound. The points $(0, 0)$, $(1, 0)$, $(2, 0)$ are pairwise at distances $1, 1, 2 \in D$, forming a triangle; hence $\chi \geq 3$.

Together $\chi = 3 \neq 7$. □

Remark. The mistake in the conjecture is presumably that the relevant “hard” instances of lattice distance graphs involve non-axis distances (e.g. $D = \{1, \sqrt{2}, 2\}$ or $\{1, 2, 5\}$ -type sets with knight moves); for the purely axis-parallel set $\{1, 2, 4\}$ every edge is axis-parallel and the mod-3 diagonal coloring applies.

3 Machine verification

`Batch9.lean` (namespace `TLMCBatch9`) contains `two_sq_ne_one` ($2x^2 \neq 1$ in $\mathbb{Z}/3$), the arithmetic core `key_arith`, the coloring `col` with correctness `col_proper`, the clique `triangle_clique`, and the packaged refutation `t1m1556` with `t1m1556_false`. Compilation via `lake env lean Batch9.lean` exits 0 with zero errors and warnings; `#print axioms` reports the standard three axioms at most (`t1m1556_false` depends on none); no `sorry` appears.